

BISBEE DOUGLAS, AZ, USA

WMO: 722720

Lat: 31.458N

Lon: 109.606W

Elev: 4105

StdP: 12.64

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 93026

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.9	25.2	-4.0	5.2	52.6	0.1	6.4	53.2	28.0	57.4	24.8	55.2	3.8	0	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	32.8	99.9	62.3	97.5	62.1	95.2	62.1	69.5	83.2	68.8	82.3	68.1	81.5	11.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	114.9	71.8	65.8	111.1	71.5	64.6	106.3	71.7	36.2	83.0	35.4	82.0	34.7	81.0	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	21.6	18.7	14.0	104.2	4.5	1.9	10.8	105.5	8.2	106.6	5.6	107.6	2.4	109.0		
(5)			11.9	72.1	4.4	1.2	8.7	73.0	6.2	73.7	3.7	74.3	0.6	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	46.4	49.8	55.6	61.6	69.9	79.3	80.3	78.4	74.7	64.9	54.2	46.1		
	DBStd	13.58	6.04	6.27	6.06	6.14	5.51	4.41	3.63	3.39	3.86	6.11	6.35	6.04		
	HDD50	423	141	72	23	3	0	0	0	0	0	2	32	150		
	HDD65	2432	577	425	292	131	20	0	0	0	1	76	324	586		
	CDD50	5360	29	68	197	352	618	880	939	881	742	465	159	30		
	CDD65	1895	0	0	2	31	173	430	474	416	293	75	1	0		
	CDH74	25266	7	54	319	1045	3136	6377	5257	4208	3288	1382	187	6		
	CDH80	12695	0	4	41	288	1483	3835	2889	2118	1544	473	20	0		
(14) Wind	WSAvg	7.7	6.7	7.8	8.7	9.8	9.4	8.8	7.6	6.4	6.8	7.2	6.8	6.6		
(15) Precipitation	PrecAvg	12.4	0.7	0.6	0.5	0.2	0.2	0.4	3.1	3.0	1.3	1.0	0.6	0.9		
	PrecMax	18.3	3.2	2.0	2.2	1.5	1.1	2.4	6.3	6.9	3.9	5.3	1.7	3.6		
	PrecMin	6.7	0.0	0.0	0.0	0.0	0.0	0.0	1.1	0.2	0.0	0.0	0.0	0.0		
	PrecStd	3.3	0.7	0.6	0.5	0.3	0.3	0.6	1.5	1.5	1.0	1.4	0.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.9	79.4	84.0	89.8	98.1	103.4	102.2	99.0	96.0	91.3	82.5	74.1		
		MCWB	49.2	50.1	52.0	54.0	59.0	61.8	64.1	65.5	63.1	60.5	53.5	49.7		
	2%	DB	70.4	75.1	80.8	86.1	94.0	100.7	99.3	96.0	93.4	88.2	79.3	70.3		
		MCWB	47.4	48.6	50.9	53.2	57.1	61.2	64.1	65.7	62.9	58.7	52.7	48.5		
	5%	DB	67.1	71.6	77.8	83.4	91.0	98.6	96.6	93.4	91.2	85.4	75.5	66.0		
		MCWB	46.1	47.3	49.8	52.2	56.2	60.7	64.1	65.6	62.6	57.9	51.8	46.6		
	10%	DB	63.3	67.4	74.4	80.7	88.3	96.1	93.6	90.8	89.0	82.2	72.1	62.2		
		MCWB	44.8	46.1	48.8	51.3	55.1	60.1	64.4	66.0	62.5	56.8	50.5	45.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.3	54.9	57.2	61.7	67.9	71.0	71.3	69.0	65.2	57.4	53.2		
		MCDB	63.5	66.3	75.7	83.0	87.7	84.6	84.0	85.2	81.6	80.1	68.4	64.3		
	2%	WB	50.2	51.1	52.8	55.2	59.7	66.0	69.5	69.9	67.9	62.8	55.4	50.8		
		MCDB	63.0	66.5	74.2	80.7	86.3	84.5	82.5	84.0	81.1	78.7	71.5	63.9		
	5%	WB	48.2	49.5	51.2	53.4	58.1	64.5	68.8	69.1	66.7	61.1	53.5	48.7		
		MCDB	62.2	66.3	73.1	79.5	85.5	85.8	81.8	83.1	80.3	77.7	70.6	62.0		
	10%	WB	46.2	47.7	49.7	51.9	56.4	63.1	68.1	68.3	65.8	59.4	51.9	46.8		
		MCDB	60.7	64.9	71.5	77.8	84.6	87.0	81.1	82.0	79.6	75.7	69.1	60.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	32.3	32.1	33.4	34.0	34.2	32.8	25.4	24.9	27.5	32.2	33.1	30.9		
		MCDBR	38.9	40.9	39.4	38.2	37.6	35.4	30.6	28.7	31.5	35.8	38.4	37.6		
		MCWBR	21.2	20.7	18.7	17.7	16.6	13.9	9.0	7.5	10.7	15.4	19.2	20.3		
		MCDBR	31.7	32.6	34.8	34.4	33.9	28.9	23.7	24.1	24.4	28.4	31.9	31.0		
	5% WB	MCWBR	18.1	16.6	16.5	16.2	15.0	10.5	6.4	6.8	8.1	12.2	16.4	17.4		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.252	0.271	0.282	0.300	0.327	0.376	0.359	0.328	0.285	0.259	0.248		
	taud		2.669	2.663	2.574	2.522	2.512	2.444	2.435	2.460	2.527	2.609	2.647	2.661		
	Ebn at Noon		318	326	323	320	312	302	287	292	299	308	312	313		
	Edh at Noon		23	26	30	33	34	36	36	35	31	27	23	22		
(44) All-Sky Solar Radiation	RadAvg		1179	1484	1946	2370	2595	2516	2087	1998	1845	1618	1295	1087		
	RadStd		83	118	112	89	97	137	109	72	82	99	66	52		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-139	+135	N/A

Nomenclature: See separate page